# Maths — Assignment - 4

1: $f(x) = x^2 + 6x + 5$ $\qquad x = 3$

$f'(x) = 2x + 6$ $\qquad \alpha = 0.1$

$$x_{k+1} = x_k - 0.1(2x_k + 6)$$
$$= 0.8x_k - 0.6$$

for $x = 3$

$x_1 = 0.8(3) - 0.6$

$= 2.4 - 0.6$

$x_1 = 1.8$

for $x = 1.8$

$x_2 = 0.8(1.8) - 0.6$

$= 1.44 - 0.6$

$x_2 = 0.84$

for $x = 0.84$

$x_3 = 0.8(0.84) - 0.6$

$= 0.672 - 0.6$

$x_3 = 0.072$

2. $f(x, y) = x^2 + 3y^2$

$\Rightarrow \nabla f = (2x, 6y)$

so, $x_{k+1} = x_k - 0.1(2x_k)$ $\qquad y_{k+1} = y_k - 0.1(6y_k)$

$\Rightarrow x_{k+1} = 0.8x_k$ $\qquad y_{k+1} = 0.4y_k$

iteration 1

$x_1 = 0.8(2) = 1.6$ $\qquad y_1 = 0.4(2) = 0.8$

iteration 2

$x_2 = 0.8(1.6)$ $\qquad y_2 = 0.4(0.8)$

$= 1.28$ $\qquad = 0.32$

iteration 3

$x_3 = 0.8(1.28)$ $\qquad y_3 = 0.4(0.32)$

$x_3 = 1.024$ $\qquad y_3 = 0.128$

so $\longrightarrow (x_1, y_1) = (1.6, 0.8)$

$(x_2, y_2) = (1.28, 0.32)$

$(x_3, y_3) = (1.024, 0.128)$.

3. $f(x,y) = x^2 y$ $\quad g(x,y) \Rightarrow x+y-10=0$

$$f(x) = x^2(10-x)$$
$$= 10x^2 - x^3$$

$$\Rightarrow f'(x) = 20x - 3x^2$$
$$f'(x) = x(20-3x) = 0$$
$$\Rightarrow x = 0 \text{ or } 20/3$$

for $x = 20/3$

$$y = 10 - 20/3 = 10/3$$

at $f''(x) = 20 - 6x$ ; $x = 20/3$

$$= 20 - 6 \times \frac{20}{3}$$
$$= 20 - 40 = -20 < 0$$

so maximum value will be at $f(20/3, 10/3)$

$$= \left(\frac{20}{3}\right) \times \frac{10}{3}$$

$$f_{max} = \frac{400}{9} \times \frac{10}{3} = \frac{4000}{27}.$$

4. $f(x,y) = x^2 + y^2 + 4x$ ; $x + 2y = 12$

$y = 6 - x/2$

$$f(x) = x^2 + (6 - x/2)^2 + 4x$$
$$= x^2 + 36 + x^2/4 - 6x + 4x$$

$$f(x) = \frac{5x^2}{4} - 2x + 36$$

$$f'(x) = \frac{5x}{2} - 2 \Rightarrow x = 4/5.$$

so, $y = 6 - 4/10 = 28/5$.

$$f_{min} = f(4/5, 28/5) = \left(\frac{4}{5}\right)^2 + \left(\frac{28}{5}\right)^2 + 4(4/5)$$

$$f_{min.} = \frac{880}{25} = 88.2$$

5. $f(x) = 4x^2 - 8x + 3$

$f'(x) = 8x - 8 \qquad f''(x) = 8$

since, $f''(x) > 0$, the function is convex

$\therefore$ Minimum: Set $f'(x) = 0 \Rightarrow 8x = 8 \Rightarrow \boxed{x = 1}$

Minimum value $f(1) = 4(1)^2 - 8(1) + 3$

$= 4 - 8 + 3 = \boxed{-1}$.

## Jacobian Part

1. $f(x) = x^3 + 2x^2 - \sin x$

$f'(x) = 3x^2 + 4x - \cos x$

2. $f(x) = \ln(x^2+1)$

$f'(x) = \dfrac{1}{x^2+1} \cdot 2x$

3. $f(x,y) = x^2y + y^3 + 2xy$

$\dfrac{\partial f}{\partial x} = 2xy + 2y \qquad \dfrac{\partial f}{\partial y} = x^2 + 3y^2 + 2x$

$\nabla f(x,y) = (2xy + 2y,\ x^2 + 3y^2 + 2x)$

4. $f(x,y,z) = x^2y + yz^2 + xz$

$\dfrac{\partial f}{\partial x} = 2xy + z \qquad \dfrac{\partial f}{\partial y} = x^2 + z^2 \qquad \dfrac{\partial f}{\partial z} = 2zy + x$

$\Rightarrow \nabla f(x,y,z) = (2xy + z,\ x^2 + z^2,\ 2yz + x)$

5. $r(t) = \begin{bmatrix} t^2 \\ \ln t \\ t\sin t \end{bmatrix} \Rightarrow r'(t) = \begin{bmatrix} 2t \\ 1/t \\ \sin t + t\cos t \end{bmatrix}$

6. $F(x, y) = \begin{bmatrix} x^2 + xy \\ y^2 + x \end{bmatrix} \begin{matrix} (f_1) \\ (f_2) \end{matrix}$

$f_1 \longrightarrow \dfrac{\partial f_1}{\partial x} = 2x + y \quad , \quad \dfrac{\partial f_1}{\partial y} = x$

$f_2 \longrightarrow \dfrac{\partial f_2}{\partial x} = 1 \quad , \quad \dfrac{\partial f_2}{\partial y} = 2y$

so, $J = \begin{bmatrix} 2x+y & x \\ 1 & 2y \end{bmatrix}$

7. $u = x^2 + y^2 \qquad v = xy$

$\dfrac{\partial u}{\partial x} = 2x \qquad \dfrac{\partial u}{\partial y} = 2y$

$\dfrac{\partial v}{\partial x} = y \qquad \dfrac{\partial v}{\partial y} = x$

so, $J = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}$ so $\dfrac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & 2y \\ y & x \end{vmatrix}$

$\Rightarrow 2(x^2 - y^2)$

8. $x = t^2 \qquad y = \sin t \qquad f(x,y) = x^2 + xy$

$f(t) = (t^2)^2 + t^2(\sin t)$

$f'(t) = (2t^2 + \sin t)(2t) + t^2 \cos t$